AI vs AI Pool Tournament Platform

v2 - Attention Architecture | April 2026

1. Concept

Competitive platform where players build AI agents to play pool. Shared open-source game engine (physics + rules). Each player's AI is a black box: game state in, shot parameters out. Tournaments with ELO ratings and prizes.

2. Game Engine

- C physics engine: sliding friction (mu=0.2) + rolling friction (mu=0.01)
- Ball collisions: elastic, multi-pass for rack propagation
- Spin: backspin (draw), topspin (follow), stop -- via angular velocity
- Cushion rebounds, pocket detection, full 14.1 rules
- Deterministic: same input = same output
- Speed: 0.056ms/shot (18,000 shots/sec single-threaded)

3. AI Interface

Input (State):
- Cue ball: (x, y)
- 15 object balls: (x, y, on_table) each
- Scores, fouls, run length, game phase

Output (Action):
- Aim angle: continuous, 0 to 2*pi
- Speed: continuous, 0 to 1
- Spin contact point: continuous, -1 (draw) to +1 (follow)

4. Reference AI: Attention Network

The Core Idea

Like an LLM uses attention over every token to predict the next word, the pool AI uses attention over every ball position to decide the next shot. Each ball is a "token." Attention weights learn which balls matter: obstructions, clusters, key balls, pocket angles -- all from self-play, no hand-coded rules.

Architecture

Input: 16 ball tokens x 12 features (position, type, pocket distances, cue distance)
Attention: 2-4 self-attention layers, 128-256 hidden dim
Global: scores, fouls, run length pooled with attention output
Actor: aim angle (Gaussian), speed (Beta), spin (Gaussian)
Critic: scalar value V(s)
Total: 500K - 2M parameters

What It Learns (no hand-coding)

- Shot selection: which ball, which pocket
- Path awareness: avoid obstructed shots
- Position play: spin + speed for cue ball control
- Safety play: when and how to play defensive
- Break ball management: key ball near rack
- Rack breaking: pocket key ball, send cue into rack
- Multi-step planning: via value function + attention
- All discovered through self-play on H100 GPU

5. Training

Hardware: NVIDIA H100 (shared with Unitree G1 project)
Parallel: 1000+ simultaneous games
Algorithm: PPO with GAE, continuous actions
Training time: 2-6 hours

Curriculum:
1. Scattered balls -- learn pocketing
2. Partial racks -- learn position + breaking
3. Full 14.1 -- complete rules
4. Self-play league -- play past versions

6. Tournament System

- 5-second compute limit per shot
- Sandboxed execution (Docker)
- ELO ratings, round-robin/bracket
- Browser spectator with shot replay
- Categories: heuristic-only, open

7. Roadmap

Phase 1 (DONE): C physics, 15K MLP on CPU, browser demo
Phase 2 (NEXT): Attention network on H100, end-to-end learning
Phase 3: Tournament platform, API, leaderboard
Phase 4: 8-ball, 9-ball, snooker support